
GENERAL RELATIVITY

Tutorial problem set 12, 04.12.2020, with solutions.

■ **PROBLEM 32** Free fall into a black hole.

Consider the Schwarzschild metric,

$$ds^2 = -A(r)dt^2 + \frac{dr^2}{A(r)} + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2), \quad (32.1)$$

where

$$A(r) = 1 - \frac{2MG_N}{r}. \quad (32.2)$$

- (a) Generally, if the metric does not depend on one of the coordinates x^α , then the momentum component, $p_\alpha = \partial\mathcal{L}/\partial\dot{x}^\alpha$ is constant along the geodesic parametrized with the affine parameter σ , $dp_\alpha/d\sigma = 0$. That follows from the Euler-Lagrange equation for the Lagrangian $\mathcal{L} = \frac{1}{2}g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}$.

Show that for Schwarzschild metric the following quantities

$$E = A(r)\frac{dt}{d\sigma}, \quad L = r^2 \sin^2\vartheta \frac{d\varphi}{d\sigma} \quad (32.3)$$

are conserved along a geodesic parametrized by σ .

- (b) Because of spherical symmetry, the free fall of a particle in Schwarzschild space-time is always confined to a plane. Choose $\vartheta = \pi/2$ (equatorial plane), and express the Schwarzschild line element as a function of E , L , $A(r)$, $dr/d\sigma$, r^2 , and $d\sigma^2$.
- (c) Show that the quantity

$$\epsilon = -g_{\mu\nu}(x)\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} \quad (32.4)$$

is constant along the geodesic (in general space-times). One has $\epsilon = 0$ for light-like trajectories, and one can choose the affine parameter σ such that $\epsilon = 1$ for a massive particle (in which case σ is the proper time τ), and $\epsilon = -1$ for space-like geodesics.

Express $(dr/d\sigma)^2$ as a function of $A(r)$, E , L , and ϵ .

Consider now an object falling radially into the black hole. Its speed at a distance $r = R$ from the center of the black hole is $dr/d\tau = U$ at $t = 0$.

- (d) Express the constant E on such a trajectory in terms of M , R and U .
- (e) Compute the 4-velocity of the falling observer as a function of the parameters E and M , and the coordinate r .
- (f) How much proper time $\Delta\tau$ does it take for the observer to reach the horizon that is located at $r = 2MG_N$? For simplicity set $E = 1$.

- (g) Argue that an external static observer sees that the object reaches the horizon in an infinite amount of time Δt .

SOLUTION: (a) That was basically shown in the lecture for the general case when we derived the Killing equation. From the midterm you know that the corresponding Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial x^\alpha} - \frac{d}{d\sigma} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} \right) = 0 \quad (32.5)$$

is equivalent to the geodesic equation if the Lagrangian is given by,

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu}(x) \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}. \quad (32.6)$$

If the metric does not depend on a coordinate x^α , then the Euler-Lagrange equation (32.5) reduces to,

$$\frac{dp_\alpha}{d\sigma} = 0, \quad p_\alpha = \frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha}. \quad (32.7)$$

One such coordinate is time, *i.e.*

$$-p_0 = A(r) \frac{dt}{d\sigma} \quad (32.8)$$

is conserved and physically it corresponds to a conserved energy per unit mass, $E = -p_0$. Furthermore, since the metric (32.1) does not depend on φ ,

$$p_\varphi = r^2 \sin^2(\vartheta) \frac{d\varphi}{d\sigma} \equiv L \quad (32.9)$$

is the conserved angular momentum per unit mass, $L = \partial_\varphi$, in the xy plane (often stated as the angular momentum vector along the z axis).

- (b) Taking account of the conserved quantities, the line element (32.1) can be rewritten as,

$$\left(\frac{ds}{d\sigma} \right)^2 = -\frac{E^2}{A(r)} + \frac{1}{A(r)} \left(\frac{dr}{d\sigma} \right)^2 + \frac{L^2}{r^2}, \quad (32.10)$$

where $d\vartheta = 0$. A convenient choice of σ is such that $ds^2 \propto d\sigma^2$, the proper time corresponding to, $ds^2 = -d\sigma^2$.

- (c) Along a geodesic we have,

$$\begin{aligned} \frac{d}{d\sigma} \left[-\frac{\epsilon}{2} \right] &= g_{\mu\nu} \frac{d^2 x^\mu}{d\sigma^2} \frac{dx^\nu}{d\sigma} + \frac{1}{2} (\partial_\alpha g_{\mu\nu}) \frac{dx^\alpha}{d\sigma} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} \\ &= -\frac{1}{2} [\partial_\alpha g_{\beta\nu} + \partial_\beta g_{\alpha\nu} - \partial_\nu g_{\alpha\beta}] \frac{dx^\alpha}{d\sigma} \frac{dx^\beta}{d\sigma} \frac{dx^\nu}{d\sigma} + \frac{1}{2} (\partial_\alpha g_{\mu\nu}) \frac{dx^\alpha}{d\sigma} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} = 0, \end{aligned} \quad (32.11)$$

where in the last line we made use of the geodesic equation.

The second part of the question is answered by (32.11) with $\left(\frac{ds}{d\sigma} \right)^2 = -\epsilon$.

- (d) For such an object, $\epsilon = 1$, and for a radial trajectory, $L = 0$ such that equation (32.11) can be rewritten as,

$$-1 = -\frac{E^2}{A(r)} + \frac{1}{A(r)} \left(\frac{dr}{d\sigma} \right)^2 \implies E = \sqrt{1 - \frac{2GM}{R} + U^2}. \quad (32.12)$$

(e) From the first equation in (32.12) we see that,

$$u^\mu = \left(\frac{dt}{d\tau}, \frac{dr}{d\tau}, 0, 0 \right), \quad \frac{dt}{d\tau} = \frac{E}{A(r)}, \quad \frac{dr}{d\tau} = \pm \sqrt{E^2 - A(r)} = \pm \sqrt{U^2 + A(R) - A(r)}, \quad (32.13)$$

where the sign should be chosen from the sign of the initial velocity, $\pm = \text{sign}[U]$.

(f) To solve for the proper time interval, we ought to integrate the radial equation in (32.13),

$$\Delta\tau(R, r_S) = \pm \int_R^{r_S} \frac{dr}{\sqrt{U^2 + A(R) - A(r)}}, \quad (32.14)$$

with $r_S = 3GM$. To make a progress, first note that if $U > 0$ then the observer will first go up, reach a maximum height (if U is smaller than the escape speed), and then plummet towards the black hole, reaching after some time again $r = R$, but now with $\frac{dr}{d\tau} = -U < 0$. If U is greater than the escape speed at $r = R$, the observer will never come back. Since this case is not really interesting to further investigate, we shall assume that $U < 0$, and therefore,

$$\Delta\tau(R, r) = - \int_R^r \frac{d\tilde{r}}{\sqrt{U^2 - \frac{2GM}{R} + \frac{2GM}{\tilde{r}}}}, \quad (32.15)$$

There are two cases to be considered separately, Case A: $\alpha \equiv U^2 - \frac{2GM}{R} > 0$ and Case B: $\beta \equiv \frac{2GM}{R} - U^2 > 0$. In Case A the integral (32.15) evaluates to,

$$I_A(r) = \frac{1}{\alpha} \sqrt{r(\alpha r + r_S)} - \frac{r_S}{\alpha^{3/2}} \text{Arcsinh} \left[\frac{2\alpha}{r_S} r + 1 \right], \quad \alpha = U^2 - \frac{2GM}{R}, \quad (32.16)$$

while in Case B the integral is,

$$I_B(r) = -\frac{1}{\beta} \sqrt{r(r_S - \beta r)} + \frac{r_S}{\beta^{3/2}} \text{Arcsin} \left[\frac{2\beta}{r_S} r - 1 \right], \quad \beta = \frac{2GM}{R} - U^2, \quad (32.17)$$

where $\text{Arcsinh}(z) = \ln(z + \sqrt{z^2 - 1})$. The proper times are then, Case A: $\Delta\tau_A(R, r_S) = I_A(r_S) - I_A(R)$ and Case B: $\Delta\tau_B(R, r_S) = I_B(r_S) - I_B(R)$. The special case asked in the problem corresponds to $E = 1$ ($U^2 = 2GM/R$), which can be obtained from as the limiting case of the above integrals. The proper time interval is in this case,

$$\Delta\tau(R, r_S) = \frac{2}{3\sqrt{r_S}} \left(R^{3/2} - r_S^{3/2} \right), \quad (32.18)$$

(g) In order to calculate the coordinate time interval, recall that $dt/d\tau = E/A(r)$. Combining this with (32.13) one obtains,

$$\Delta t = -E \int \frac{dr}{A(r)\sqrt{E^2 - A(r)}} = - \int \frac{dr}{1 - \frac{r_S}{r}} \sqrt{\frac{r}{r_S}}, \quad (32.19)$$

where the first equality holds for a general E , while the second for the special case, $E = 1$. For simplicity, here we just calculate the special case. A convenient variable to use is $x = \sqrt{r/r_S}$, in terms of which the indefinite integral in (32.19) evaluates to,

$$I_C(x) = -2r_S \left[\frac{x^3}{3} + x + \frac{1}{2} \ln \left(\frac{x-1}{x+1} \right) \right]. \quad (32.20)$$

Inserting this into (32.19) we obtain for the time interval,

$$\Delta t(R, r) = -2r_S \left[\frac{1}{3} \left(\frac{r}{r_S} \right)^{\frac{3}{2}} + \left(\frac{r}{r_S} \right)^{\frac{1}{2}} + \frac{1}{2} \ln \left(\frac{\left(\frac{r}{r_S} \right)^{\frac{1}{2}} - 1}{\left(\frac{r}{r_S} \right)^{\frac{1}{2}} + 1} \right) \right] + [\text{the same with } r \rightarrow R]. \quad (32.21)$$

In the limit when $r \rightarrow r_S$, the log term contributes as,

$$\approx -r_S \ln \left[\frac{1}{4} \left(\frac{r}{r_S} - 1 \right) \right],$$

which – close to r_S – dominates and diverges logarithmically. This means that, from the point of view of an asymptotic observer at rest (for which t is the proper time), it will take infinitely long for an object to fall into the black hole. This can be explained by an infinite time dilation at the event horizon.

■ PROBLEM 33 Gravitational redshift in Schwarzschild space-time.

Consider two observers in Schwarzschild space-time, at fixed spatial coordinates $(r_1, \vartheta_0, \varphi_0)$ and $(r_2, \vartheta_0, \varphi_0)$ outside of the Schwarzschild radius $r_S = 2GM$ (in this problem assume $c = 1$). Suppose that observer 1 (let's call him Marko) emits a light signal radially towards observer 2 (let's call him Bruno).

- (a) How are the proper times of these observers related to the coordinate time t ?

Label the proper time interval (as measured by Marko) between the emission of two light crests by $\Delta\tau_1$. What is the relation between $\Delta\tau_1$, and Δt , where Δt is the interval of coordinate time between the emission of two light crests (you can view Δt as the time elapsed between the emission of two crests as measured by an observer at $r \rightarrow \infty$).

- (b) What is the proper time interval $\Delta\tau_2$ (as measured by Bruno) between the two successive crests received? Express your result in terms $\Delta\tau_1$, r_S , r_1 and r_2 .
- (c) Compute the ratio of the two frequencies ν_1 (emitted by Marko), and ν_2 (received by Bruno), and show that in the limit $r_1, r_2 \gg r_S$ one has,

$$\frac{\nu_2}{\nu_1} = 1 + \phi(r_1) - \phi(r_2), \quad (33.1)$$

where $\phi(r_1)$ and $\phi(r_2)$ are the gravitational potentials at the respective coordinates.

SOLUTION: (a) The relation between the proper and coordinate times at the observer 1 (Marko) and 2 (Bruno) are,

$$d\tau_1 = \sqrt{1 - \frac{r_S}{r_1}} dt, \quad d\tau_2 = \sqrt{1 - \frac{r_S}{r_2}} dt. \quad (33.2)$$

The coordinate time interval dt is the proper time interval of asymptotic observers at rest at $r \rightarrow \infty$; while $d\tau_1$ and $d\tau_2$ are the proper time intervals measured by observers 1 and 2, respectively. Since the radius is fixed to $r = r_1$, we can integrate Eq. (33.2) to obtain,

$$\Delta\tau_1 = \sqrt{1 - \frac{r_S}{r_1}} \Delta t, \quad (33.3)$$

(b) Observer 2 will measure,

$$\Delta\tau_2 = \sqrt{1 - \frac{r_S}{r_2}} \Delta t = \sqrt{\frac{1 - \frac{r_S}{r_2}}{1 - \frac{r_S}{r_1}}} \Delta\tau_1 = \sqrt{\frac{g_{00}(r_2)}{g_{00}(r_1)}} \Delta\tau_1. \quad (33.4)$$

(c) The times intervals measured between two crests correspond to measurement of a change in phase $\Delta\Phi(x) = \omega\Delta t$ of a wave, which is an invariant for all observers. For an asymptotic observer we have, $\Delta\Phi = \omega_0\Delta t$, while for observers 1 and 2,

$$\Delta\Phi = \omega_1\Delta\tau_1 = \omega_2\Delta\tau_2 \quad (33.5)$$

where ω_1 and ω_2 are the frequencies measured by observers 1 and 2, respectively, and ω_0 is the frequency measured by the asymptotic observer. From this we conclude,

$$\frac{\omega_1}{\omega_2} = \frac{\Delta\tau_2}{\Delta\tau_1} = \sqrt{\frac{1 - \frac{r_S}{r_2}}{1 - \frac{r_S}{r_1}}} = \sqrt{\frac{g_{00}(r_2)}{g_{00}(r_1)}}. \quad (33.6)$$

This means that the wave will have higher frequency (energy) deeper in the gravitational potential. In particular, when a wave of frequency ω_E emitted at r climbs out of a potential well it will exhibit a gravitational redshift which amounts to,

$$\frac{\omega_0}{\omega_S} = \sqrt{1 - \frac{r_S}{r}} = \sqrt{-g_{00}(r)} < 1, \quad (33.7)$$

where ω_S is the emitted frequency and ω_0 is the observed frequency by the asymptotic observer.

More generally, g_{00} can be written as, $g_{00}(r) = 1 + 2\phi(r)$, where ϕ is the gravitational potential. Inserting this into (33.6) and expanding to linear order in ϕ yields,

$$\frac{\omega_1}{\omega_2} \simeq 1 + [\phi(r_2) - \phi(r_1)] \implies \frac{\omega_2}{\omega_1} = \frac{\nu_2}{\nu_1} \simeq 1 + [\phi(r_1) - \phi(r_2)] . \quad (33.8)$$

which is what we were supposed to show.